Cost Function / Gradient Descent

Name _____ Date _____

01 Cost Function

Shared Data

Input x	-4	-3	-2	-1	1	2	3	4
Target y	-3	-2	-1	0	2	3	4	5

1 Keep $b = 0$ and **fill in the blanks**

$$f_w(x) = wx \qquad (m = 8)$$

$$\text{MSE} = \frac{1}{m}\sum_{i=1}^{m}\left(f_w(x^{(i)}) - y^{(i)}\right)^2$$

$$J(w) = \frac{1}{2m}\sum_{i=1}^{m}\left(f_w(x^{(i)}) - y^{(i)}\right)^2 = \frac{\text{MSE}}{2} = \frac{1}{16}\sum_{i=1}^{8}\left(wx^{(i)} - y^{(i)}\right)^2$$

* MSE (Mean Squared Error) = the mean of squared errors

	Prediction $f_w(x)$									
w	$x=-4$	$x=-3$	$x=-2$	$x=-1$	$x=1$	$x=2$	$x=3$	$x=4$	MSE	$J(w)$
-1	4	3	2	1	-1	-2	-3	-4	31	15.5
0	0	0	0	0	0	0	0	0	8.5	4.25
1	-4	-3	-2	-1	1	2	3	4	1	0.5
2	-8	-6	-4	-2	2	4	6	8	8.5	4.25
3	-12	-9	-6	-3	3	6	9	12	31	15.5

2 For $w = 1$, substitute the eight errors to **calculate** $J(1)$

$$\text{Error}^{(i)} = f_1(x^{(i)}) - y^{(i)}$$

$$J(1) = \frac{(\underline{-1})^2 + (\underline{-1})^2 + (\underline{-1})^2 + (\underline{-1})^2 + (\underline{-1})^2 + (\underline{-1})^2 + (\underline{-1})^2 + (\underline{-1})^2}{16} = \underline{0.5}$$

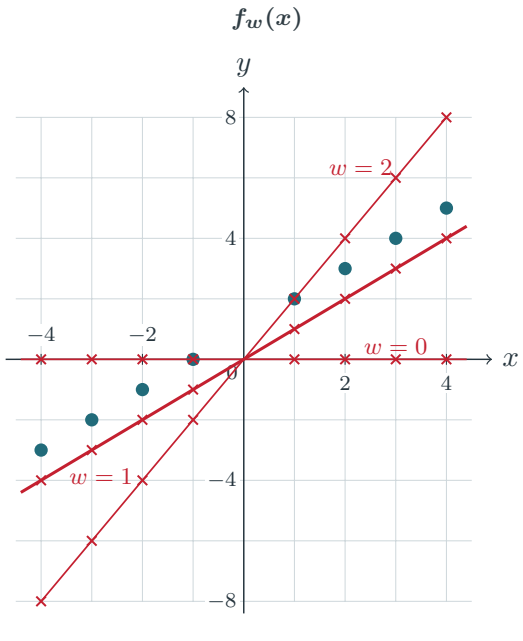
Reference · DeepLearning.AI / Stanford Online, Machine Learning Specialization, C1 W1, slides 40 to 57 · Original examples and questions

02 Predictions and Cost

Use the table on page 1

3 **Draw the lines** for $w = 0, 1, 2$
Label each line with w

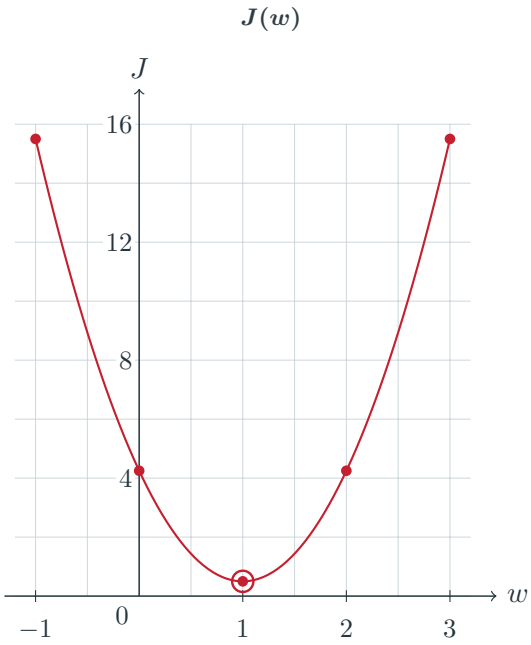
$$f_w(x) = wx \quad (b = 0)$$



• Target

4 Plot the five points $(w, J(w))$
Connect them with a smooth curve

$$J(w) = \frac{1}{16}\sum_{i=1}^{8}(wx^{(i)} - y^{(i)})^2$$



5 Trace the $w = 1$ line on the left and circle its corresponding cost point on the right

6 Write either x or w in each blank

Input supplied in the data _____ x _____

Value changed to choose a prediction line _____ w _____

7 Among the five candidates, find **the** w **with the lowest cost** and that cost

$w =$ _____ 1 _____ $J(w) =$ _____ 0.5 _____

Explain whether this line passes through all eight actual data points

It does not pass through all eight points

Every prediction is 1 below its target

03 Changing the Bias

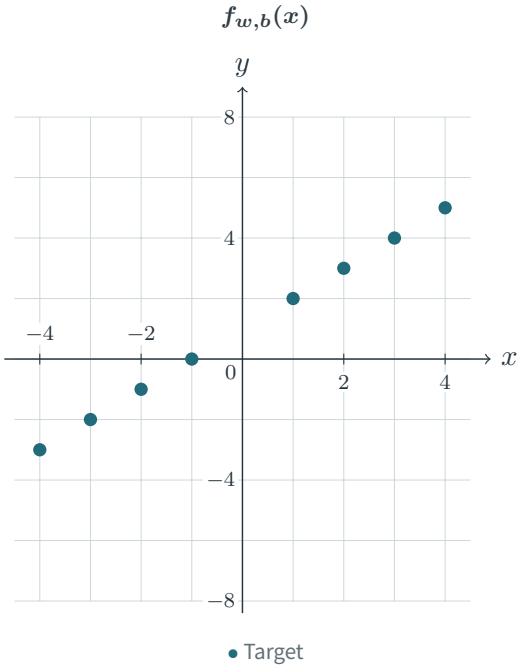
8 **Calculate the cost** of the nine lines

$$f_{w,b}(x) = wx + b \qquad (m = 8)$$

$$J(w,b) = \frac{1}{16}\sum_{i=1}^{8}\left(wx^{(i)} + b - y^{(i)}\right)^2 = \frac{\text{MSE}}{2}$$

* w = Weight, b = Bias

			Prediction $f_{w,b}(x)$									
Name	w	b	$x=-4$	$x=-3$	$x=-2$	$x=-1$	$x=1$	$x=2$	$x=3$	$x=4$	**MSE**	$J(w,b)$
A	0	0	0	0	0	0	0	0	0	0	8.5	4.25
B	1	0	−4	−3	−2	−1	1	2	3	4	1	0.5
C	2	0	−8	−6	−4	−2	2	4	6	8	8.5	4.25
D	0	1	1	1	1	1	1	1	1	1	7.5	3.75
E	1	1	−3	−2	−1	0	2	3	4	5	0	0
F	2	1	−7	−5	−3	−1	3	5	7	9	7.5	3.75
G	0	2	2	2	2	2	2	2	2	2	8.5	4.25
H	1	2	−2	−1	0	1	3	4	5	6	1	0.5
I	2	2	−6	−4	−2	0	4	6	8	10	8.5	4.25



9 Describe how the **predictions change** from B to E to H

$$f_{1,b}(x) = x + b \quad (b = 0, 1, 2)$$

Every prediction increases by 1 at each step

10 Find **the equation of the zero-cost line** and its predictions at both endpoints

$$f_{w,b}(x) = wx + b$$

$f_{w,b}(x) =$ $x + 1$

$f_{w,b}(-4) =$ -3, $f_{w,b}(4) =$ 5

Choose two lines with equal cost

A and C

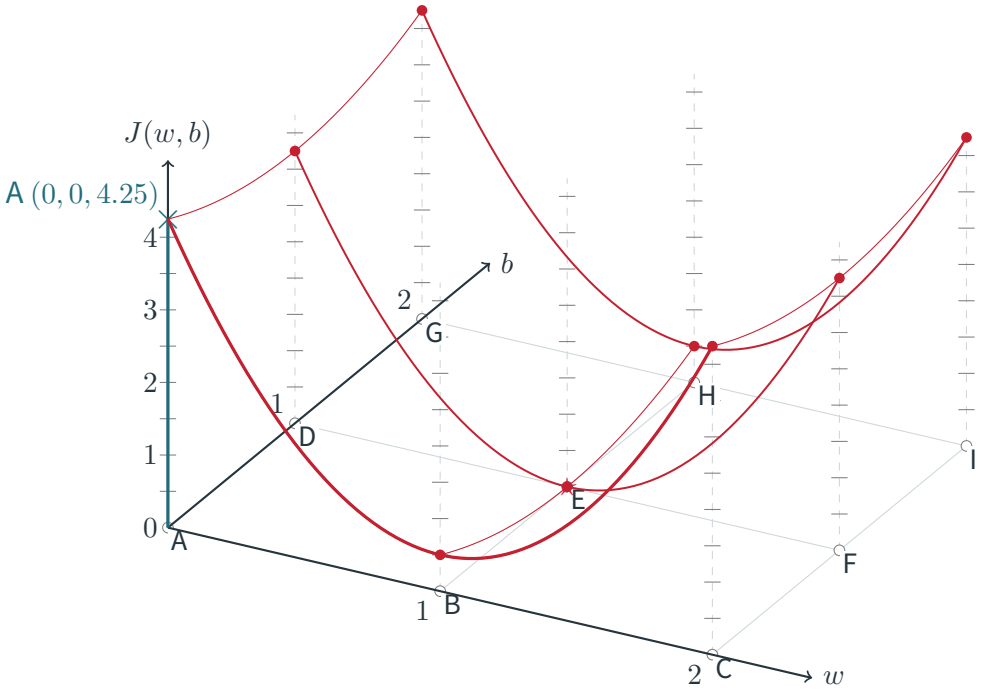
04 Cost Surface

11 Use the table on page 3 to **complete the coordinates** and plot the other eight points

B $(1, 0, \underline{0.5})$ C $(2, 0, \underline{4.25})$ D $(0, 1, \underline{3.75})$

E $(1, 1, \underline{0})$ F $(2, 1, \underline{3.75})$ G $(0, 2, \underline{4.25})$

H $(1, 2, \underline{0.5})$ I $(2, 2, \underline{4.25})$



12 Connect A→B→C, D→E→F, and G→H→I with smooth curves

Value held fixed along each curve $\underline{b}$

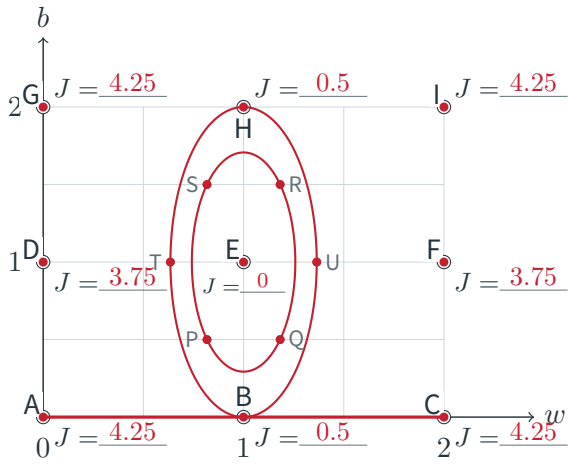
13 Lightly connect A→D→G, B→E→H, and C→F→I
Mark the lowest point with a star and write its coordinates

$(w, b, J(w, b)) = (\underline{1}, \underline{1}, \underline{0})$

05 Contour Plot

14 Write the costs at A through I and **draw two contours**

$$J(w,b) = \frac{1}{16}\sum_{i=1}^{8}\left(wx^{(i)} + b - y^{(i)}\right)^2$$



Cost = 0.5
Connect B, U, H, and T with a smooth closed curve

Cost = 0.25
Connect P, Q, R, and S in the same way

15 Trace the $b = 0$ row and find the same curve on the 3D diagram on page 4

16 Compare the **minimum costs** for the two choices

$$\min_{w} J(w) \quad \text{and} \quad \min_{w,b} J(w,b) \quad \big(J(w) = J(w,0)\big)$$

Free parameters	Best w	Its b	Min. cost
Choose w; fix $b = 0$	1	0	0.5
Choose both w, b	1	1	0

17 Explain how changing b changes the set of lines you can choose

$$f_w(x) = wx \qquad f_{w,b}(x) = wx + b$$

With $b = 0$, every line passes through the origin

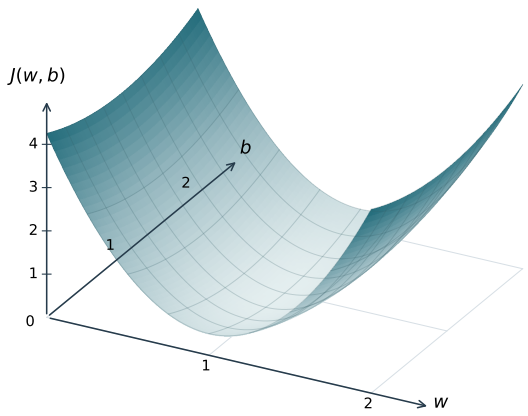
Changing b shifts a line up or down, allowing lines that do not pass through the origin

06 Choosing a 3D Cost Surface

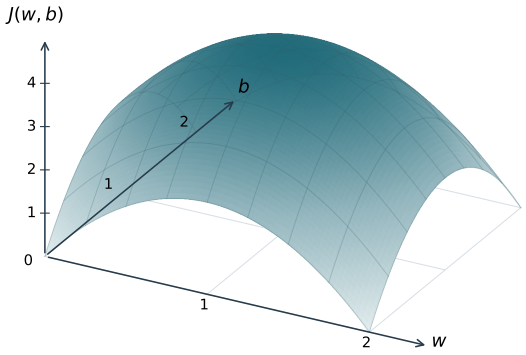
18 Choose the **3D cost surface** that matches the shape you drew on page 4

$$f_{w,b}(x) = wx + b, \qquad J(w,b) = \frac{1}{16}\sum_{i=1}^{8}\left(wx^{(i)} + b - y^{(i)}\right)^2$$

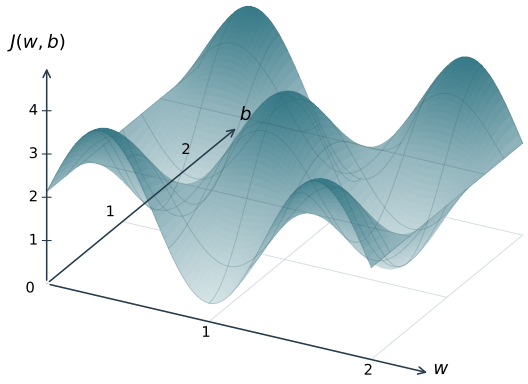
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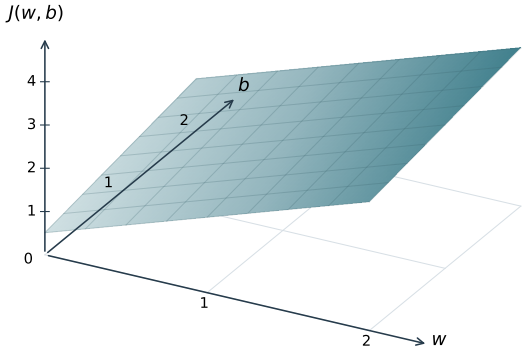
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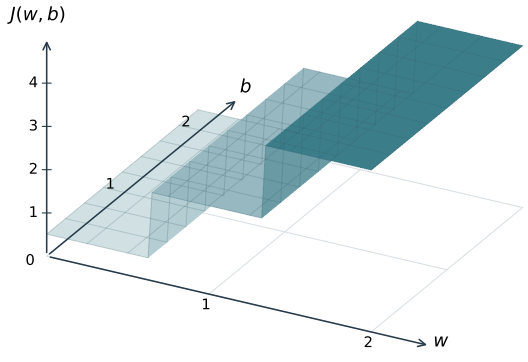
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④



⑤



Selected number 1

Explain your choice

$J(w,b) = \frac{15}{4}(w-1)^2 + \frac{1}{2}(b-1)^2$

The minimum is 0 at $w = b = 1$

Bowl-shaped: cost rises away from the minimum